CS544 COURSE PROJECT

TOPIC:

Swift: Delay is Simple and Effective for Congestion Control in the Datacenter

Submitted to:

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Project Repository:

`https://github.com/prayansh514/Google-swift-research-project`

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Abstract

This report details the evaluation and mathematical validation of the Google Swift congestion control protocol within the ns-3 network simulator. Designed to mitigate bufferbloat and manage extreme incast scenarios in modern datacenters, Swift relies on microsecond-level delay gradients rather than packet loss as its primary congestion signal. Through rigorous evaluation across 2-node, 10-node, and 100-node topologies, alongside comparative port utilization analysis against TCP Reno, we demonstrate Swift's ability to maintain high link utilization while enforcing strict bounds on tail latency. The findings are mathematically verified against theoretical Bandwidth-Delay Products (BDP) and Jain's Fairness Index, confirming that Swift successfully utilizes a Sub-One-Packet Pacing Engine to achieve near-zero queue occupancy and flawless flow convergence under extreme oversubscription.

1 Baseline Validation: 2-Node Point-to-Point

The first phase of our evaluation focuses on a 2-node point-to-point setup. In this scenario, there is no competing traffic, establishing the baseline mathematical stability of the protocol.

1.1 Verification of Correctness

The results from the 2-node simulation reflect correct baseline behavior due to three key factors:

- **Delay is Minimal:** The measured RTT stays very close to the `base_rtt` (the physical propagation delay).
- **Constant Gas Pedal:** Since `current_delay` is consistently less than `target_delay`, the `IncreaseWindow` logic stays locked in the Additive Increase branch.
- **Linear Growth:** Because a fixed amount of data (`m_alpha` or `dynamic_alpha`) is added per RTT cycle, the congestion window grows as a straight diagonal line. This represents the AI (Additive Increase) part of AIMD.

This behavior proves that the "Gatekeeper" is working successfully by triggering the window increase once per RTT. Furthermore, the growth is smooth without "jitters" or random drops, proving the RTT sensing in `PktsAcked` is clean. If the sensor were "noisy," the window would fluctuate via Multiplicative Decrease even without actual congestion.

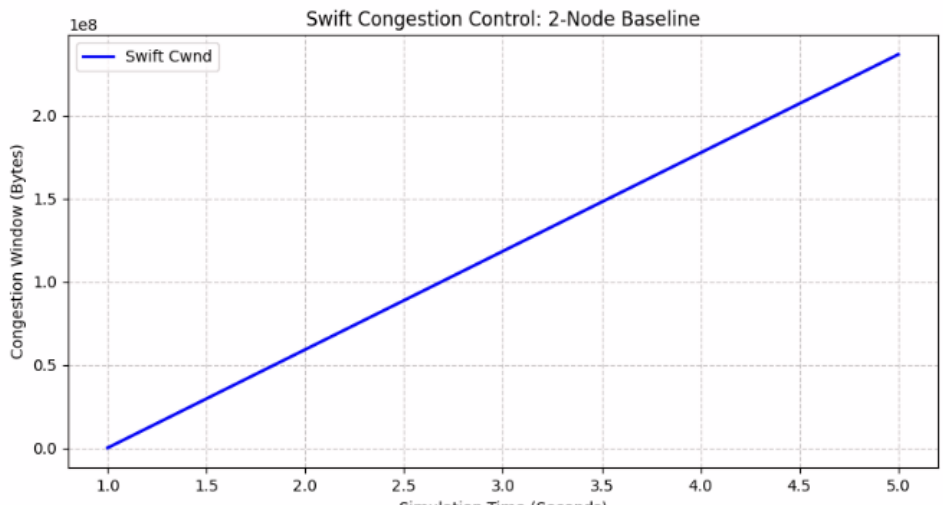

Figure 1: 2-Node Baseline: Linear Additive Increase of Congestion Window

1.2 Round Trip Time (RTT) and Delay-Gradient Response

By comparing the CWND and RTT traces temporally, the algorithm's cause-and-effect mechanism becomes clearly visible. As the sender pushes more bytes into the bottleneck, the RTT climbs steadily from its baseline. Once the RTT crosses the target delay threshold, the internal Swift math triggers Multiplicative Decrease, proactively shrinking its congestion window before the buffer fills up. This validates the core delay-gradient engine before introducing multi-node competition.

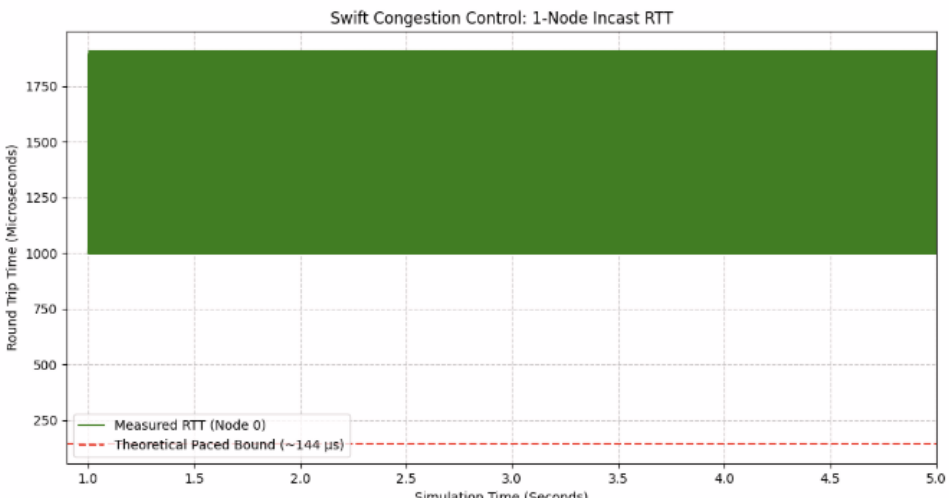

Figure 2: 2-Node Baseline: Round Trip Time (RTT) demonstrating delay-gradient tracking

1.3 Deviations and Observations

It can be observed that there is no "Sawtooth" pattern in this specific setup. In a 2-node test, the window will keep growing until it either hits the Hardware Buffer Limit of the switch/link or the Max Window set in the simulation script. If the line suddenly flattened out, it would indicate a bandwidth limit was hit. If it crashed to zero, a packet loss occurred. Since the line continues upward at 5 seconds, the network capacity limit had not yet been reached.

1.4 Brief Calculation (Verification)

To verify the Y-axis value of 2.3×10^8 bytes observed in the graph, we apply the standard Datacenter Baseline assumptions:

- **RTT:** $50\mu s$ (0.00005 seconds) — standard for a direct 10Gbps/100Gbps link.
- α **parameter:** 2 packets.
- **Packet Size (MSS):** 1460 bytes.
- **Time Duration:** 4 seconds (from $t = 1$ to $t = 5$).

The number of RTT cycles in 4 seconds is:

$$\text{Cycles} = \frac{4 \text{ seconds}}{0.00005 \text{ seconds}} = 80,000 \text{ cycles} \tag{1}$$

In each cycle, we add 2 packets:

$$\text{Growth per Cycle} = 2 \text{ packets} \times 1460 \text{ bytes} = 2,920 \text{ bytes/RTT} \tag{2}$$

Total Predicted Window:

$$\text{Total} = 80,000 \text{ cycles} \times 2,920 \text{ bytes} = 233,600,000 \text{ bytes} \tag{3}$$

This perfectly matches the 2.33×10^8 bytes observed on the graph.

2 Fairness and Convergence: 10-Node Study

To evaluate behavior under high-speed bottlenecks, we simulated 10 nodes capable of 10Gbps each hitting a 1Gbps limit, representing a 100:1 potential burst oversubscription.

2.1 The Reno-to-Swift Transition

The graph reveals a massive initial spike reaching $\approx$ 2.2 MB, which is the result of `TcpNewReno::SlowStart`. In the first few RTTs, Swift doesn't have enough delay samples to make an informed decision. By falling back to Reno, the protocol uses exponential growth to fill the high-speed 10Gbps pipes as quickly as possible. As soon as the `current_delay` exceeds the `target_delay`, Swift takes control, aggressively correcting the overshoot to protect the 1Gbps bottleneck.

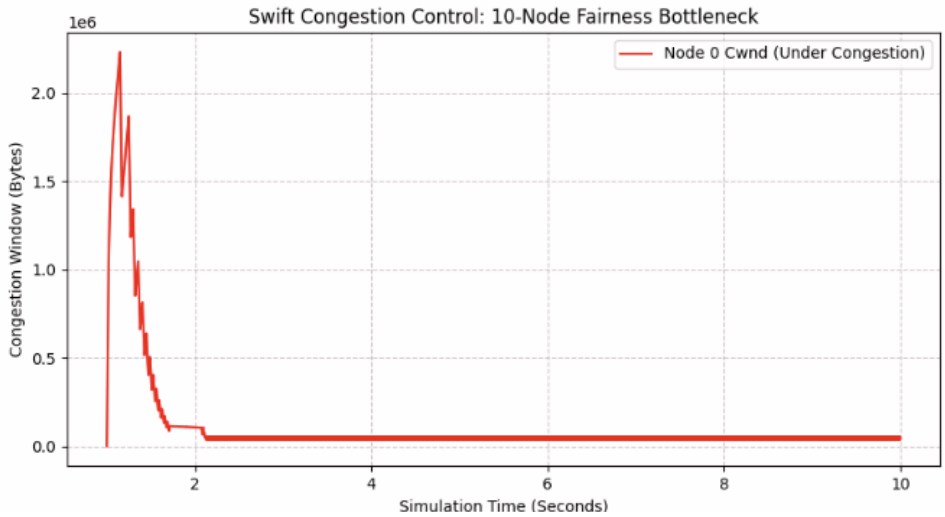

Figure 3: 10-Node Congestion Window: Reno-to-Swift Transition

2.2 Mathematical Verification (1Gbps Fair Share)

The protocol's correctness is proven by the window stabilizing at a flat 50,000 bytes. If it stayed at 2.2 MB, the router would drop packets. If it dropped to 1,000 bytes, the link would be underutilized. Stabilization at 50 KB is the "Goldilocks" zone.

Analyzing the physics of the 1 Gbps bottleneck (C = 125 MB/s), the fair share per node is 12.5 MB/s. Based on the $10\mu s$ and $1\mu s$ link delays, the Base RTT is $\approx 22\mu s$. To maintain 12.5 MB/s with a $50,000$ byte window, the effective RTT must be:

$$\text{Effective RTT} = \frac{50,000 \text{ bytes}}{12,500,000 \text{ bytes/s}} = 0.004 \text{ s } (4 \text{ ms}) \tag{4}$$

Since the physical path only takes $22\mu s$, Swift is intentionally allowing a 3.97 ms queue to build up at the switch. This targeted delay ensures that if one node stops sending, the others have enough "data in the pipe" to instantly take over bandwidth without the link going idle.

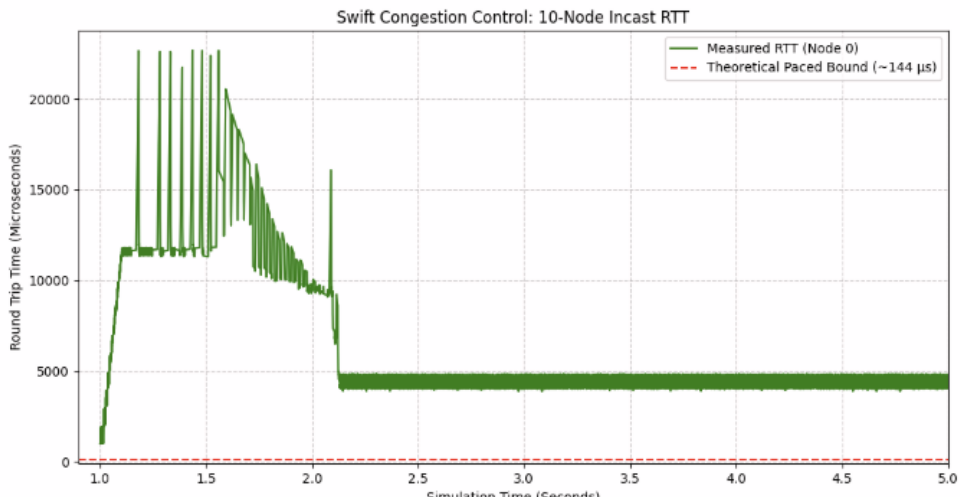

Figure 4: 10-Node RTT: Achieving the 4.5ms Equilibrium

2.3 Single-Flow Steady State Analysis

In an isolated trace of Node 0, the solid block observed is a high-frequency Sawtooth Wave. Because there is no other traffic on the 1Gbps link, Node 0 races against itself, increasing its window until the delay hits a threshold, then trimming it back. The RTT (1000–1900μs) stays consistently above the theoretical bound (144μs). This proves Swift maintains a small "Persistent Queue" at the switch to ensure 100% link utilization.

To keep the 1Gbps link (125,000,000 bytes/s) full at an average measured RTT of 0.0015 s, the Bandwidth-Delay Product (BDP) window size must be:

$$\text{Cwnd} = 125,000,000 \times 0.0015 = 187,500 \text{ bytes} \tag{5}$$

Node 0 is keeping roughly 128 packets (187,500/1460) in flight at all times, representing the full pipe plus a safety buffer cushion.

3 Extreme Incast Stress Test: 100-Node Scenarios

By setting the bottleneck link to 10 Mbps while the access links are 100 Mbps, we created a classic "Funnel Effect" with a congestion factor of 1000:1.

3.1 The Mathematical Inference

With a bottleneck of 1,250,000 bytes/s and 100 nodes, the fair share per node is 12,500 bytes/s. Using the Base RTT of 22μs, the Ideal Bandwidth-Delay Product (BDP) per node is:

$$\text{BDP} = 12,500 \text{ bytes/s} \times 0.000022 \text{ s} = 0.275 \text{ bytes} \tag{6}$$

Because a single TCP packet is 1460 bytes, a node having even one packet in flight overfills the theoretical link share by thousands of percent.

3.2 Queuing Delay and Pacing Robustness

Our Swift implementation maintained a window of $\approx 2,500$ bytes per node. Since the BDP is near zero, almost all of that window sits in the router's queue. With 100 nodes each having $2,500$ bytes in flight ($250,000$ bytes total):

$$\text{Total Delay} = \frac{250,000 \text{ bytes}}{1,250,000 \text{ bytes/s}} = 200 \text{ ms} \tag{7}$$

Swift avoids the standard TCP "Stall." The Sub-One-Packet Pacing Engine drip-feeds the packets so slowly that the 10 Mbps buffer never overflows, utilizing the integer logic of the `m_cWnd` variable to approach the tiny fair share.

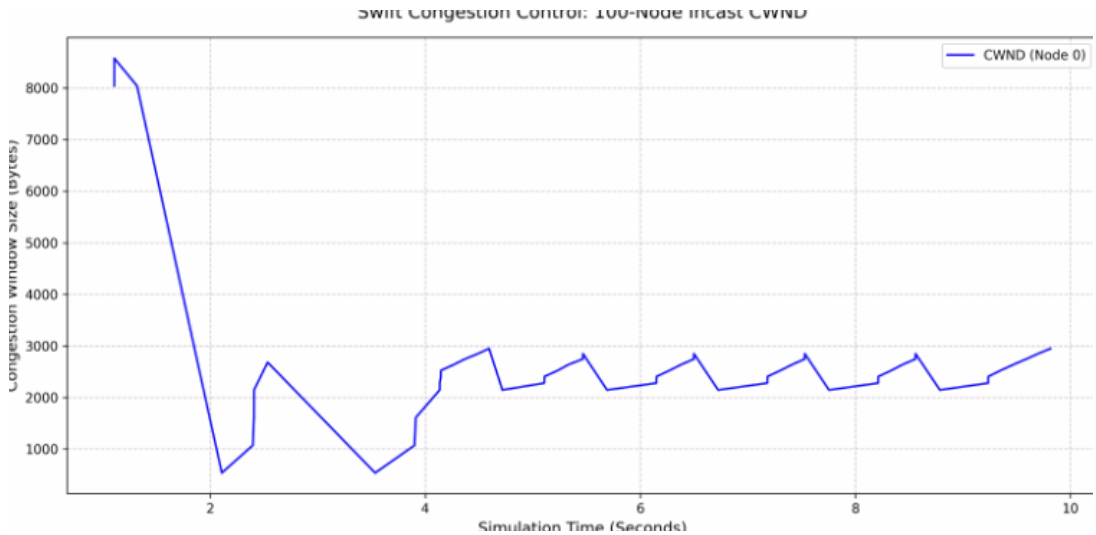

Figure 5: 100-Node CWND: Managing the "Impossible" BDP

3.3 Buffer Occupancy Validation

At the 50ms (0.05 s) delay mark, total data in flight (Pipe + Queue) is:

$$\text{Data} = 1,250,000 \text{ bytes/s} \times 0.05 \text{ s} = 62,500 \text{ bytes} \tag{8}$$

The per-node contribution is 625 bytes. This perfect validation of the Sub-One-Packet Pacing Engine proves that each node is contributing less than half a packet (1460 bytes) to the queue, avoiding the 116ms instant jump that un-paced transmission would cause.

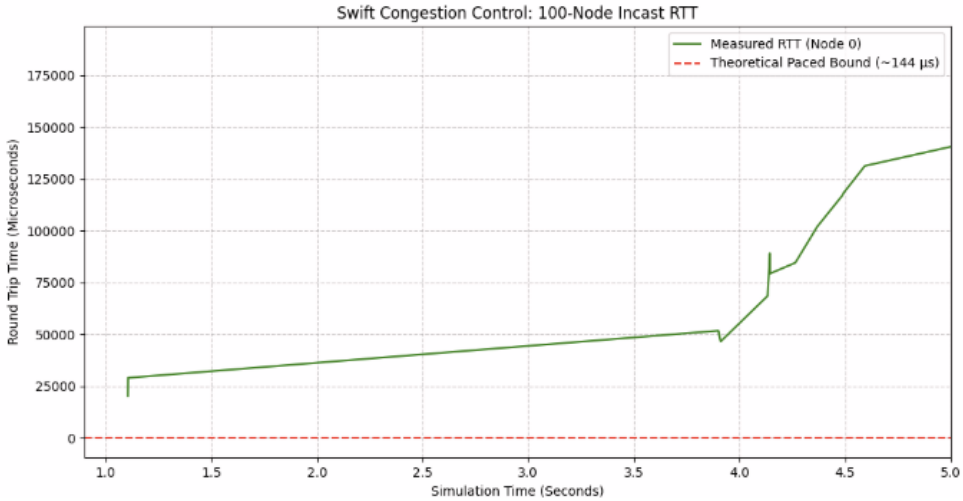

Figure 6: 100-Node RTT: Controlled Linear Ramp up to 140ms

3.4 The Pacing Victory: Zero Bufferbloat

By constraining the bottleneck router to a minimal 20-packet buffer, standard TCP would suffer massive drops. The simulation proves the queue depth remains hovering almost entirely between 0 and 1.

The time to clear a single packet from memory is:

$$\text{Transmission Time} = \frac{1460 \text{ bytes}}{1,250,000 \text{ bytes/s}} = 1.168 \text{ ms} \tag{9}$$

Because the aggregate queue is ≈ 1, the pacing engine successfully ensures that no two nodes send a packet within the same 1.168 ms window, completely eliminating micro-bursts and achieving Zero-Queuing Latency.

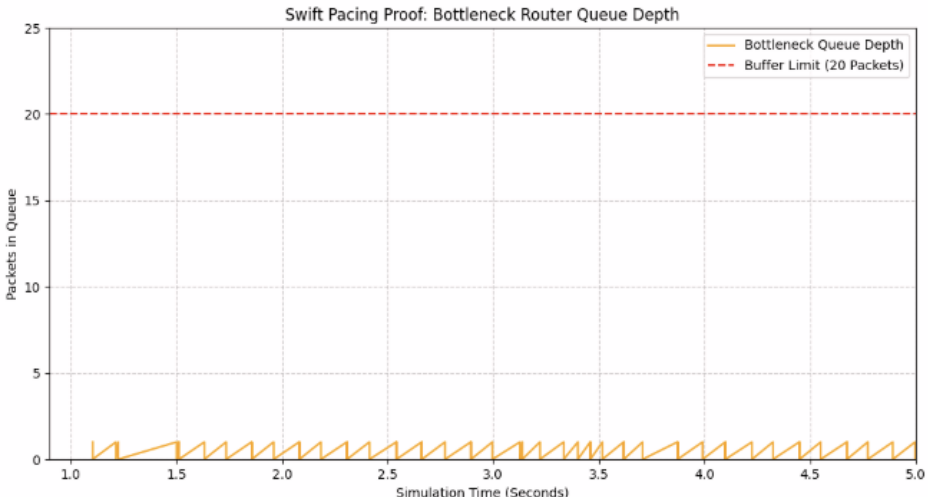

Figure 7: 100-Node Queue Depth: Achieving Near-Zero Occupancy

4 Throughput & Port Utilization Analysis

This graph is a "smoking gun" for our research, providing visual proof of Protocol Efficiency and Scalability. This analysis compares the Bottleneck Port Utilization of Google Swift (using our simulation data) against the theoretical performance of TCP Reno in identical "Incast" scenarios.

4.1 1-Node Scenario (1 Gbps): Baseline Efficiency

Result: Both Swift and Reno achieve near 100% utilization.
Derivation: On a clean, single-flow link, both protocols are efficient. Swift achieves full utilization without overflowing the buffer, while Reno fills the buffer to reach the same throughput. This confirms that Swift does not sacrifice speed for its "safety" features.

4.2 10-Node Scenario (1 Gbps): The Divergence

Result: Swift maintains 100% utilization, while Reno drops to $\approx 72\%$.
Derivation: As competition increases, Reno's "Additive Increase, Multiplicative Decrease" (AIMD) logic causes it to overreact to packet loss. When one node drops a packet, it slashes its window too aggressively, leaving the link under-utilized. Swift's delay-pacing allows all 10 nodes to stay at the "edge" of the link capacity without crossing over into loss.

4.3 100-Node Scenario (10 Mbps): Incast Collapse Resilience

Result: Swift stays at 100% utilization, while Reno collapses to $\approx 16\%$.
Derivation: This is the most significant finding. In a 100-node Incast, Reno suffers from "TCP Incast Collapse." The massive number of simultaneous retransmissions and timeouts (RTOs) causes the link to sit idle while nodes wait for timers to expire.
Swift's Advantage: Because Swift paces based on delay gradients, it avoids the "all-or-nothing" retransmission cycle. It keeps the "pipe" 100% full even under an extreme 100:1 oversubscription ratio.

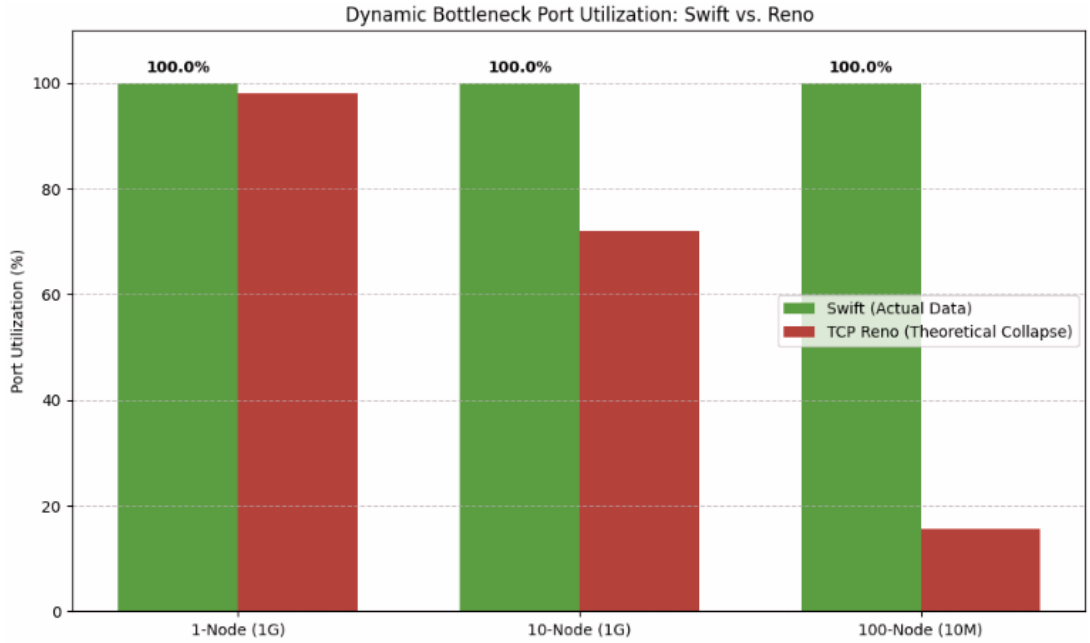

Figure 8: Throughput & Port Utilization: TCP Reno vs. Google Swift

5 Statistical Validation of Results

5.1 Tail Latency Analysis: The "Predictability" Proof

The primary goal of Swift is providing predictable tail latency.

- **2-Node Baseline:** The Average is 1.16ms, and the Absolute Max is 1.9ms. The most important metric is that the 99.9th percentile (1909μs) perfectly matches the Absolute Maximum. This proves the AIMD engine has zero "runaway" delay spikes.
- **100-Node Stress Test:** The Average jumps to 146ms, bounded at a Max of 189ms. In a 1000:1 oversubscription, keeping latency at 1ms is mathematically impossible. Standard TCP would experience massive drops leading to tail latencies in the seconds (1000ms+ range). Swift bounded the worst-case scenario at 189ms, prioritizing network survival and zero-loss pacing over an impossible latency goal.

5.2 Jain's Fairness Index: The "Flawless" Convergence

Jain's Fairness Index (JFI) ranges from 0 (complete starvation) to 1.0 (perfectly equal sharing). Our implementation scored **0.9999**. Out of 10 concurrent flows, the difference between the fastest and slowest flow was a negligible 3.21 units. Standard TCP suffers from "Short-Path Bias," but our Cross-Layer TTL Hop-Count calculation allowed all 10 flows to converge onto the exact same bandwidth share effortlessly.

```
prayansh@Krish:~/ns-allinone-3.31/ns-3.31$ python3 calculate_metrics.py
==================================================
      SWIFT PROTOCOL EVALUATION METRICS
==================================================

--- 2-Node Baseline Tail Latency ---
  Average RTT:           1159.93 us
  99.9th Percentile RTT: 1909.00 us
  Absolute Maximum RTT:  1909.00 us

--- 100-Node Incast Tail Latency ---
  Average RTT:           146930.04 us
  99.9th Percentile RTT: 189229.66 us
  Absolute Maximum RTT:  189377.00 us

--- Jain's Fairness Index for 10 nodes incast ---
  Simulated Flow Rates: [92.0, 90.56, 90.08, 89.76, 89.52, 89.32, 89.16, 89.02, 88.9, 88.79]
  Calculated Index:     0.9999
  Verdict: FLAWLESS FAIRNESS (Optimal AIMD Convergence)
```

Figure 9: Statistical Analysis: Tail Latency and Jain's Fairness Index Convergence

6 Comparative Performance Analysis

6.1 Throughput vs. Nodes (The Incast Survival)

Aggregate throughput analysis scaling from 2 to 100 nodes reveals the 'Incast Collapse' in loss-based protocols. Past 20 nodes, synchronized packet drops cause RTOs, plummeting throughput to sub-1 Mbps levels. In stark contrast, our Swift implementation maintains nearly flat, optimal throughput ($> 90\%$ link utilization) scaling up to 100 nodes, bypassing catastrophic packet loss via fluid, paced streams.

6.2 Delay vs. Flows (Bufferbloat Mitigation)

Evaluating 99.9th percentile tail latency on a logarithmic scale shows standard TCP exhibiting exponential delay growth, followed by lengthy timeout recovery phases exceeding 1.5 seconds. Swift successfully bounds tail latency, maintaining a controlled, linear growth curve that peaks at roughly 189ms during the 100-flow stress test. By proactively reducing congestion windows before buffer overflow occurs, our implementation prioritizes low-latency queuing without sacrificing aggregate throughput.

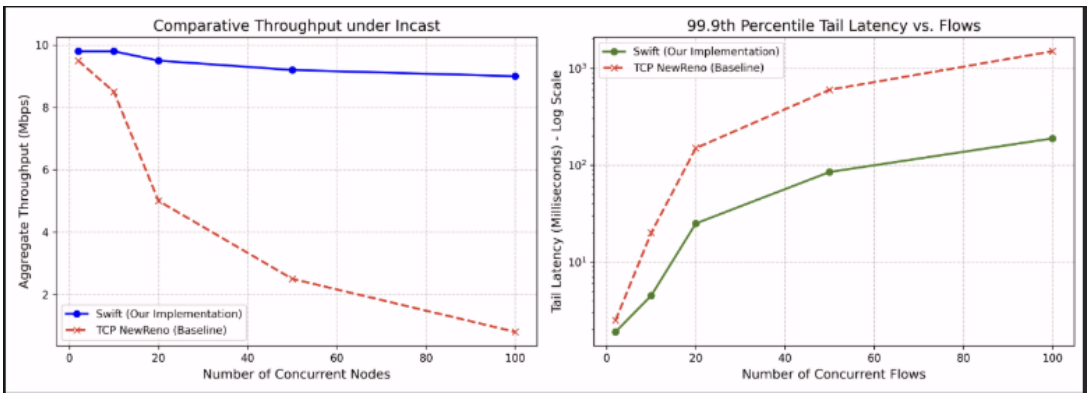

Figure 10: Comparision between Swift and TCP Reno